

Technical Summary:

Chain-of-Thought Prompting Elicits Reasoning in Large Language Models

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1 Research Problem and Motivation

Large language models perform strongly on many natural language tasks yet often struggle on problems that require multi-step reasoning in arithmetic, commonsense, and symbolic domains. Rationale-augmented training and task-specific fine-tuning improve such tasks but incur substantial data and engineering cost. This study asks whether multi-step reasoning can be elicited in sufficiently large models purely through prompt formatting, with no parameter updates, by adding exemplars that contain explicit intermediate reasoning.

2 Related Work

Rationale supervision demonstrates benefits from exposing intermediate steps on math word problems and related tasks, while few-shot prompting enables in-context learning without gradient updates. The present work combines these lines by augmenting few-shot exemplars with natural-language chains of thought, aiming to capture the benefits of rationales without training-time annotation.

3 Dataset Construction

The evaluation spans three families of tasks:

- **Arithmetic:** GSM8K, SVAMP, ASDiv, AQuA, MAWPS. These require multi-step quantitative reasoning from textual descriptions.
- **Commonsense:** CommonsenseQA (CSQA), StrategyQA, BIG-bench Date Understanding, BIG-bench Sports Understanding, and SayCan.
- **Symbolic:** Last Letter Concatenation and Coin Flip, designed to test rule following and generalization to lengths beyond the exemplar distribution.

Exemplars. Eight chain-of-thought exemplars are used when context permits. AQuA uses four multiple-choice exemplars. For CSQA and StrategyQA, exemplars come from training data. For BIG-bench subsets without training splits, the first ten evaluation items are used as exemplars and the remainder for testing. SayCan uses six training examples from prior work.

Table 1: Summary of benchmarks, exemplar policies, and evaluation metrics.

Category	Datasets	Exemplars	Primary metric
Arithmetic	GSM8K, SVAMP, ASDiv, AQuA, MAWPS	8 CoT exemplars (AQuA uses 4)	Solve rate or accuracy
Commonsense	CSQA, StrategyQA, Date, Sports, SayCan	8 exemplars; BIG-bench uses first 10 items; SayCan uses 6	Task-appropriate accuracy
Symbolic	Last Letter, Coin Flip	8 exemplars; in-domain and length OOD splits	Solve rate

4 Query Protocol and Task Definitions

Standard few-shot prompting presents input-output pairs. Chain-of-thought prompting appends an explicit reasoning sequence to each exemplar. The prompt consists of triples

$$\langle x, r, y \rangle, \quad (1)$$

where x is the input instance, r is the natural-language chain of thought that decomposes the problem into intermediate steps, and y is the final answer. Equation (1) specifies the exemplar format shown to the model. The goal is to induce the model to follow a similar intermediate reasoning trajectory when predicting on a new x .

5 Modeling Approach

Model roster and decoding

Off-the-shelf families are evaluated without fine-tuning: GPT-3 up to 175B parameters, LaMDA up to 137B parameters, PaLM at 8B, 62B, and 540B parameters, UL2 20B, and Codex. Greedy decoding is used. For LaMDA, results are averaged over random seeds that permute exemplar order.

Ablation variants

Three controlled variants test alternative hypotheses on GSM8K: *Equation only* requires emitting only an equation before the answer; *Variable compute only* requires emitting a sequence of periods whose count matches the equation length; *Chain of thought after answer* provides the chain of thought only after answering. The former two match standard prompting, indicating that natural-language intermediate steps matter; the latter also aligns with baseline, suggesting that sequential reasoning during inference contributes beyond knowledge activation.

Robustness

Gains hold across annotators, exemplar sets, exemplar counts, and multiple model families. Prompt order variance is generally small, with Coin Flip as an exception due to its classification format.

Terminology

Chain of Thought (CoT) denotes a coherent sequence of intermediate natural-language steps that lead to an answer. **Solve rate** is the percentage of correctly solved items under each benchmark’s scoring rule. **Emergent ability** refers to a capability that appears only beyond a specific model scale threshold.

6 Empirical Results

Notation. In the result tables below, **Std.** denotes standard prompting and **CoT** denotes chain-of-thought prompting. Values are taken directly from the paper without rounding.

6.1 Arithmetic reasoning

With PaLM 540B, chain-of-thought prompting establishes new state of the art on GSM8K, SVAMP, and MAWPS, and improves ASDiv and AQuA. Exact values appear in Table 2.

Table 2: Arithmetic benchmarks for PaLM 540B: standard vs chain-of-thought. Solve rate (%).

Method	GSM8K		SVAMP		ASDiv		AQuA		MAWPS	
	Std.	CoT	Std.	CoT	Std.	CoT	Std.	CoT	Std.	CoT
PaLM 540B	17.9	56.9	69.4	79.0	72.1	73.9	25.2	35.8	79.2	93.3

6.2 Commonsense reasoning

With PaLM 540B, chain-of-thought prompting achieves **77.8%** on StrategyQA, outperforming the prior single-model best of **69.4%**, and reaches **95.4%** on Sports Understanding, surpassing the unaided human baseline of **84%**. Additional results are given in Table 3.

Table 3: Commonsense benchmarks for PaLM 540B: standard vs chain-of-thought. Solve rate (%).

Method	CSQA		StrategyQA		Date		Sports		SayCan	
	Std.	CoT	Std.	CoT	Std.	CoT	Std.	CoT	Std.	CoT
PaLM 540B	78.1	79.9	68.6	77.8	49.0	65.3	80.5	95.4	80.8	91.7
Prior best / baseline	–		69.4		–		84 (human)		–	

6.3 Symbolic reasoning and length generalization

Chain-of-thought prompting yields very high solve rates on Last Letter Concatenation and Coin Flip and supports generalization to inputs longer than those shown in the exemplars. Standard prompting degrades on longer sequences, whereas CoT improves with scale.

6.4 Qualitative error analysis

Manual inspection shows that most correct answers are accompanied by logically coherent chains of thought. Scaling from PaLM 62B to 540B reduces one-step-missing and semantic understanding errors, consistent with scale-dependent emergence.

7 Summary

Innovation. A prompt-only protocol elicits multi-step reasoning in large language models by presenting chain-of-thought exemplars, removing the need for rationale-augmented fine-tuning and addressing limitations of standard prompting.

Limitations. Generated chains are not guaranteed to reflect internal computations and can be partially correct. Benefits emerge at large model scales, which increases serving cost. Scaling annotation to training size would be expensive.

Future directions. Promising directions include improving factuality and calibration of generated reasoning, transferring CoT benefits to smaller models, and studying how exemplar design influences robustness across tasks and distributions.